

Effects of visual field, exposure time, and set size on icon search with varied delays using an LCD monitor

Kuo-Chen Huang
Wen-Te Chang
Wan-Li Wei

Abstract — This study investigated the effects of visual field (left or right), exposure time, and set size on icon search with varied delay times. Nineteen women and 17 men, who were Taiwanese graduate students aged 24–40 years, participated. Participants were told to search for a target icon in a circular stimulus array, which had a diameter of 7 cm and included five, seven, or nine items. The results indicated that set size significantly affected search time. Search time was significantly longer for a nine-icon set than for five and seven icons. Search times were significantly longer for 5-sec delays than for 3- and 1-sec delays. However, exposure time and visual field did not significantly affect search time. Results showed that set size significantly affected accuracy. The accuracy for the five-icon condition was greater than that for the seven- and nine-icon conditions. Moreover, accuracy was greater for 1- than for 3- and 5-sec-delay conditions. However, no significant effect on accuracy was found for visual field or exposure time. Finally, search times for correct and incorrect responses were analyzed separately. The results are applicable to graphics-based interface design.

Keywords — Exposure time, set size, icon, search performance.

DOI # 10.1889/JSID18.6.427

1 Introduction

Icons are believed to be more useful for communicating ideas than words because of their ability to transcend language barriers and present meaning in a condensed form.¹ Icons are used in computer-graphics interfaces and in handheld devices such as PDAs.² They can be designed for hugely variable presentation conditions, such as icon location and layout or number of icons presented simultaneously at a Web site or display. This affects the speed and accuracy by which displayed icons can be searched and perceived by users.^{3,4} Therefore, icons need to be designed to optimize search times, considering the factors likely to slow a visual search.⁵ As with graphical information, the efficiency with which computer icons are searched should depend on several different factors, such as the number of icons presented simultaneously, icon location, and duration of icons' presentation to users.

1.1 Visual field

The subjects' search performance is degraded when they are forced to monitor locations with noise. Greater probability that a random noise configuration at one location will appear like a signal string is associated with subjects' mistakenly choosing the noise in a higher proportion of trials.⁶ Results of Seamon *et al.*⁷ showed that a lateralized stimulus presentation significantly affected recognition, with performance for a right-visual-field presentation superior to that for a left-visual-field presentation. Additionally, Jonsson and Hellige⁸ asked subjects to indicate as quickly and accurately as possi-

ble whether two uppercase letters were the same or different. Their results showed that when the letter pairs were projected to the right visual field, a moderate amount of blurring produced a significantly impaired performance; however, the same level of blurring did not result in impairment when the letter pairs were projected to the left visual field. Moreover, Pedersen and Polich⁹ assessed hemispheric processing differences by presenting square matrices that varied in size and the number of filled-in cells. The results indicated that response time was shorter and error rates were lower for left-visual-field presentations than right-visual-field presentations. Moreover, larger matrices demonstrated the strongest visual-field effects. These studies seem to indicate that spatial uncertainty has a significant impact on recognition.

1.2 Exposure duration

The exposure effect is defined as the increase in positive effect that results from the repeated presentation of unfamiliar stimuli.¹⁰ Previous studies have shown that visual stimulus recognition improves with increased exposure duration.^{11,12} For example, Shapiro and Penrod¹¹ reported that exposure duration significantly affects face-recognition memory. Reynolds and Pezdek¹² demonstrated that participants exposed to 20-sec stimulus duration remembered faces better than did participants exposed to 3-sec duration, implying that additional exposure time resulted in quantitative but not qualitative changes in memory processing. Furthermore, Bundesen and Harms¹³ suggested that an exponential distribution function under longer exposures well approxi-

Received 12/15/2009; accepted 03/24/2010.

The authors are with Ming Chuan University, Department of Product Design, 5 Teh-Ming Rd., Gwei-Shan, 333 Taoyuan, Taiwan; telephone +886-9338-28048, e-mail: kuochen@mcu.edu.tw.

© Copyright 2010 Society for Information Display 1071-0922/10/1806-0427\$1.00.

mated the probability (p) of recognizing the stimulus at a given exposure duration (t). Longer exposure durations might produce bidirectional interference,¹⁴ resulting in reduced identification time for the local features of a stimulus pattern.¹⁵ However, shorter exposure durations might primarily affect local features perception,¹³ resulting in a unidirectional interference pattern¹⁶ and leading to an increase in participants' difficulty recognizing the stimulus. These results showed that exposure duration plays an important role in pattern recognition, but the degree of exposure effect might be affected by discrepancies in the quality of the local and global features present in a stimulus pattern¹⁵ or might be different when experiments are conducted under various interactions among independent variables.¹⁷

1.3 Matrix size

During a visual search task, the reaction time for correctly detecting a target is measured as a function of the number of distracter elements.^{6,18,19} Lindberg and Nasanen¹⁹ investigated the effect of stimulus and set size on search speed and showed that search time increases with increasing set size. Additionally, Elmes *et al.*²⁰ suggested that a visual search is likely to be influenced by the number and spacing of stimuli in the visual field. Similarly, Verghese and McKee⁶ analyzed psychometric functions of set size and suggested that the probability of finding the target decreases as the number of locations monitored increases. Furthermore, Lavie *et al.*²¹ manipulated working memory load during a visual search and found that working memory load affects attentional capture by a salient singleton distracter during the visual search. However, Logan²² failed to find any effect of working memory load on the slopes of search-set size functions. Popple *et al.*²³ asked participants to pinpoint the precise location of a target under various search-set sizes (*e.g.*, 5×5 and 9×9 Gabor patch arrays) by presenting an odd-one-out stimulus that popped up from the background, but its location was slightly ambiguous for examining error size. Their results showed that more errors occurred with the sparse array (5×5) than with the dense array (9×9), suggesting that the effect of set size on search performance is restricted by the target and distracter characteristics presented in a search array.

1.4 Delay

Delay time is another important factor that facilitates or impedes target search and recognition. Previous studies have shown a negative relationship between delay and recognition accuracy for unfamiliar pictorial stimuli.^{24–26} These delay effects suggest that subjects use the delay between the stimulus and test to flexibly adjust the decision criteria for stimulus recognition.²⁷ Hanari²⁵ asked participants to identify the locations of two of six small dots presented for 50, 200, or 350 msec at random locations following a recall delay of 50, 250, or 500 msec after stimulus termination.

The results showed that performance deteriorated from the 50-msec to the 500-msec recall delay for the shorter exposure duration; however, no effect of delay on recall accuracy was observed for longer exposure duration. Podd²⁴ examined the effects of memory load over three periods of delay and found that increased target load decreased accuracy mainly by decreasing hit rate, and increasing delay decreased accuracy largely as a result of an increased rate of false alarm. This result implied that increasing delay reduces maintenance of visuospatial information.²⁸ These findings showed that delay is the primary factor limiting recall performance.²⁶

However, Laughery²⁹ examined the effect of delay on target recognition by manipulating the delay between seeing the target and starting the test series, and the results indicated that delay had no effect on recognition performance. Additionally, Yarmey,³⁰ using a 20-min retention interval, and Goldstein and Chance,³¹ using a 48-hour retention interval, found no effect of delay on recall for facial stimuli. Similarly, Berry and Carpenter³² examined the effect of four different delay periods (15, 30, 45, or 60 min) on recall of the Rey–Osterrieth complex figure (ROCF) in the elderly. The results showed no significant differences between the groups on immediate or delayed recall trials, indicating that the delay did not have a significant effect on recall. These results are supported by the suggestions of MacLin *et al.*³³

To summarize, set size in a search array, exposure duration, visual field, and delay are four important factors that affect search performance, and these have previously been studied separately. Therefore, we investigated the effects of these four factors on visual search simultaneously. The aim of this study was to examine whether the visual field, exposure time, and set size affect search performance and whether these interact with time delays.

2 Method

2.1 Participants

Nineteen female and 17 male graduate students between the ages of 24 and 40 years (mean $\pm$ SD, 31.8 ± 4.5 years) served as participants. All participants reported that they searched for information on the Internet at least once per week for more than 1 hour. All participants reported 16/20 corrected visual acuity or better. The participants received a reward of NT 500 dollars for their participation.

2.2 Stimulus materials and design

The four independent variables were set size in a search array, right or left visual field, exposure time, and time delay to respond. The levels of each variable are described below. The set size was defined as the number of icons in a circular search array. The three set-size levels were five, seven, and nine icons.

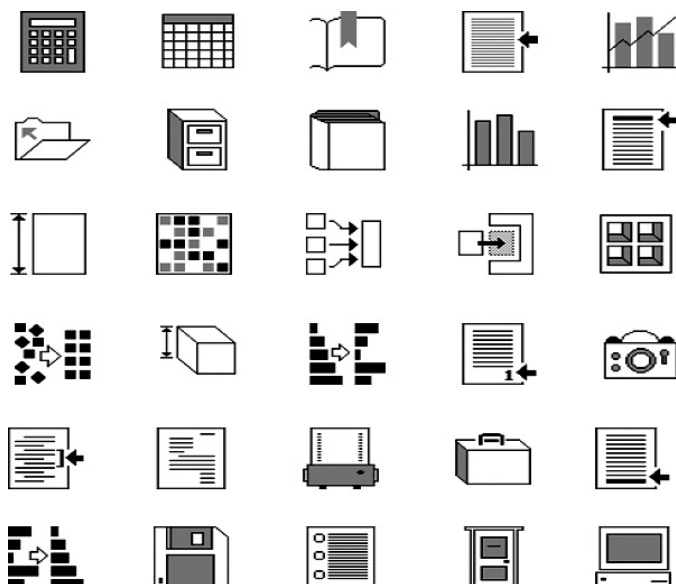


FIGURE 1 — Thirty icons were used in the present study.

The monitor screen was divided equally into right and left regions. The screen was divided so that 33% and 67% of the stimuli were presented on the left and right regions or on the right and left regions, respectively. The former was defined as the right visual field, the latter was defined as the left visual field.

Stimuli were presented with three exposure times of 1.0, 2.0, and 3.0 sec. Additionally, three levels of delay were used for the subjects to respond: 1.0, 3.0, and 5.0 sec.

Thirty icons were used (Fig. 1). Each icon was 42 pixels in height by 42 pixels in width (1 pixel = 0.32 mm). Each stimulus contained several icons depending upon the set-size independent variable. The icons for each trial were randomly chosen from the 30 icons. Thus, the observed field might contain five, seven, or nine icons. The space between two icons was kept equal within an observational field.

The icons were arranged in a 7.0-cm-diameter circle with one target icon and the other distracters (Fig. 2). The circumference of the circular array was divided equally into several regions, depending on the number of icons in the stimulus. A target could be located at any region on the circumference, and participants were not required to scan in a strict sequence.

A 3 (set size) $\times$ 2 (visual field) $\times$ 3 (exposure time) $\times$ 3 (delay) $\times$ 30 (trials for each set size level) mixed design was used. The exposure time and delay were between-subjects variables, whereas the set size and visual field were within-subject variables. Each experimental run had three blocks of 30 trials. A single block included 20 stimuli displayed in the right and 10 in the left visual field, or vice versa. The visual field variable was balanced and randomized across participants. Moreover, set size for each stimulus was held constant for each block. The interval between blocks was 1 min. The stimuli, designed using Flash 2004 MX, were produced by a visual simulator and displayed in the center of a Slim

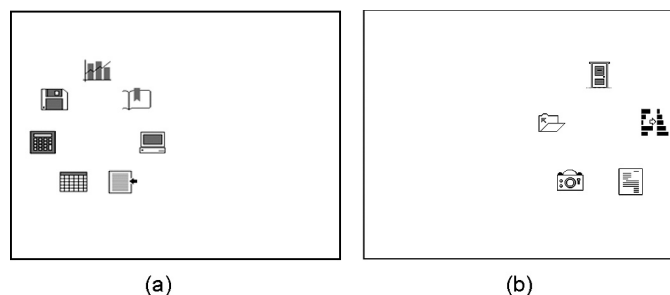


FIGURE 2 — Two examples of circular search array for experiment: (a) seven icons displayed in the left field; (b) five icons displayed in the right field. Within a search array, it holds the same spacing between icons.

Age 17-in LCD monitor with Pentium-four PC-based Microsoft software.

2.3 Procedure

Participants first completed a practice series consisting of five trials, followed by a series of 90 experimental trials, and the order of trials for each participant was randomized. For both the practice and experimental trials, participants were instructed as follows: “Click START using the mouse to produce and focus on the cross. After the cross disappears, memorize the target icon presented on the screen. After the target icon disappears, the stimulus array will immediately be presented on the right or left side of the screen. You should search for the target, move the mouse to the target item, and press the mouse button. Your search time and accuracy will be recorded, so please pay careful attention while performing this procedure.” All participants were tested individually in a computer classroom in which the light level was approximately 300 lux. To ensure maximum clarity and replicability, the contrast was set to the maximum level and brightness to a minimum level. The viewing distance was approximately 50 cm.

At the beginning of each trial, a fixation point appeared in the center of the screen for 1 sec. Subjects were instructed to fixate visually on this point before the presentation of each stimulus. A target icon was presented once the fixation point disappeared. Subjects were instructed to memorize the target icon as quickly and as accurately as possible. After the target icon disappeared, the screen displayed a “blank” interval lasting several seconds, depending on the time delay. Then, a circular search array was presented on the right or left side of the screen. Participants had to search for the target icon in the stimulus array, move the mouse to the target, and press the button as quickly and as accurately as possible. The search time and accuracy during this process were recorded by computer. The experiment took approximately 45-minutes.

2.4 Data analysis

Search performance was analyzed in terms of speed (seconds) and accuracy (percentage of errors) using a repeated-

measures analysis of variance, which was conducted using SPSS software (SPSS, Inc., Chicago, IL, U.S.A.). Only the mean performance data for the icon search under different experimental conditions were collected because the effect of icon type on performance was not considered in the present study.

3 Results

3.1 Accuracy

Entries in Table 1 are means and standard errors. Results showed a significant main effect of set size on accuracy ($F_{2,54} = 4.78, p < 0.05, \eta_p^2 = 0.15$). Duncan's test showed that the accuracy for the five-icon condition was significantly greater than that for the seven- and nine-icon conditions. However, no difference was found between the seven- and nine-icon conditions. The results also showed main effects for delay on accuracy ($F_{2,27} = 3.98, p = 0.03, \eta_p^2 = 0.23$). Duncan's test demonstrated that the accuracy was significantly greater for 1-sec conditions than that for the 3- and 5-sec conditions. However, no difference was found between the 3- and 5-sec conditions. No significant effect on accuracy was associated with visual field or exposure time, $F_{1,27} = 0.51, p = 0.48, \eta_p^2 = 0.02$; $F_{2,27} = 0.12, p = 0.59, \eta_p^2 = 0.01$, respectively. No two-way interactive effect on accuracy was found among these four independent variables.

3.2 Search time

The results indicated a significant main effect of set size on search time ($F_{2,54} = 14.99, p < 0.001, \eta_p^2 = 0.36$). Duncan's test showed that search time for the nine-icon condition was significantly longer than that for the five- and seven-icon conditions. However, the difference in search time between the five- and seven-icon conditions was not significant. An

analysis of variance showed that the main effects for delay on search time were significant ($F_{2,27} = 5.64, p < 0.01, \eta_p^2 = 0.30$). Duncan's test demonstrated that search time was significantly longer for 5 sec than that for 1- and 3-sec conditions. However, no difference was observed between the 1- and 3-sec conditions. An analysis conducted on data accuracy revealed no differences attributable to visual field ($F_{1,27} = 3.09, p = 0.09, \eta_p^2 = 0.10$) or exposure time ($F_{2,27} = 0.8, p = 0.79, \eta_p^2 = 0.02$), and reliable two-way interactions were observed among these four variables.

Search time data for correct and incorrect response were also analyzed separately. First, the correct response data were analyzed in an identical ANOVA with four independent variables. The search time data for correct responses showed no effect of set size ($F_{2,54} = 0.27, p = 0.76, \eta_p^2 = 0.01$), visual field ($F_{1,27} = 1.03, p = 0.32, \eta_p^2 = 0.04$), exposure time ($F_{2,27} = 2.23, p = 0.13, \eta_p^2 = 0.14$), or delay ($F_{2,27} = 0.46, p = 0.64, \eta_p^2 = 0.03$). A two-way interaction between set size and delay for the correct search time was found ($F_{4,54} = 4.24, p < 0.01, \eta_p^2 = 0.24$; Fig. 3). The simple effect on correct search time of set size (five *vs.* seven *vs.* nine) was significant for the 5-sec delay (1.98 *vs.* 2.10 *vs.* 2.48, respectively; $F_{2,22} = 7.89, p < 0.01, \eta_p^2 = 0.42$); however, this effect was not significant for a delay of 3-sec ($F_{2,22} = 3.04, p = 0.07, \eta_p^2 = 0.22$) or 1-sec ($F_{2,22} = 2.31, p = 0.12, \eta_p^2 = 0.17$).

Second, a repeated-measure ANOVA with set size and visual field as within-subject variables was performed for incorrect response search time. The main effect of set size on incorrect search time was significant ($F_{2,54} = 7.31, p < 0.01, \eta_p^2 = 0.21$). A Duncan's test revealed shorter incorrect search time for both the five- (mean $\pm$ SE, 2.04 ± 0.09) and seven-icon (2.17 ± 0.10) than the nine-icon (2.52 ± 0.10) condition, but showed no difference between the five- and seven-icon conditions. The incorrect response data revealed no effect of visual field ($F_{1,27} = 0.1, p = 0.76, \eta_p^2 = 0.01$), exposure time ($F_{2,27} = 2.31, p = 0.12, \eta_p^2 = 0.15$), and delay ($F_{2,27} = 3.29, p = 0.06, \eta_p^2 = 0.20$), and a significant two-way interactive effect was found among these four variables.

TABLE 1 — Means and standard errors of accuracy and search time under each level of the independent variables ($N = 36$).

	Accuracy (%)			Search time (sec)		
	<i>M</i>	<i>SE</i>	Duncan test	<i>M</i>	<i>SE</i>	Duncan test
Set size						
5	93.54	2.63	B	1.92	0.04	A
7	85.80	2.78	A	2.06	0.07	A
9	85.97	2.07	A	2.34	0.06	B
Visual field						
Left	89.22	2.56	A	2.07	0.04	A
Right	87.65	1.73	A	2.14	0.04	A
Exposure time						
1	88.54	3.28	A	2.12	0.07	A
2	87.24	3.28	A	2.07	0.07	A
3	89.53	3.28	A	2.13	0.07	A
Time of delay						
1	95.91	3.28	B	2.02	0.07	A
3	85.62	3.28	A	2.02	0.07	A
5	83.78	3.28	A	2.29	0.07	B

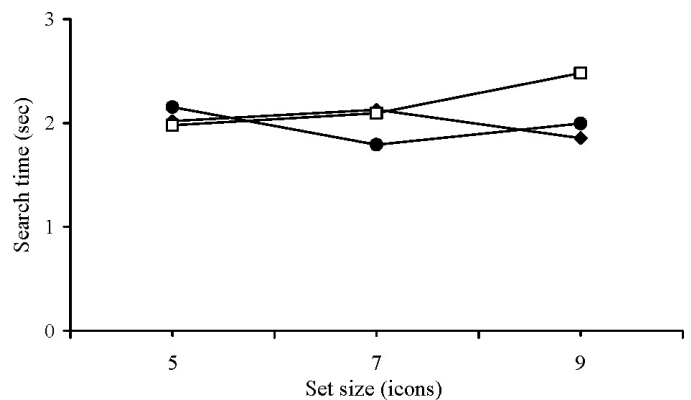


FIGURE 3 — Search time for correct response as a function of set size and delay (◆ — 1 sec; ● — 3 sec; □ — 5 sec).

4 Discussion

Set size significantly affected search performance, and the nine-icon set size yielded lower accuracy and longer search time than did the five-icon size. This result was consistent with those obtained by Lindberg and Nasanen¹⁶ and similar to the suggestions of Vlaskamp *et al.*¹⁸ and Treisman and Gelade.³⁴ Two explanations can be offered for the results. First, the search for a target icon was serial^{16,34} in the present study, as the target had a conjunction of features and could not be detected based on any single feature; it did not pop up, but instead required attention before it could be found among the distracters. When a target icon does not pop up from distracters, subjects take a longer time to bind more icon features with a larger set size because the binding must be more robust in the presence of more potential distracters.³⁵ Additionally, the concept of perceptual load implies that more icons are added for the same task, and a more demanding perceptual task is conducted under a higher perceptual load.²¹ Therefore, when more icons are presented in the search array, subjects might use more attentional capacity to search for a target icon that facilitates distracter discrimination, and this might result in an increase in the interference effects of distracters during such a search task. Second, a decrease in search performance for increased set sizes may result from crowding¹⁸ under a given diameter of circular stimulus array, making it more difficult for subjects to search for a target icon that is surrounded by other icons in close proximity. The inter-icon distance among nine icons was smaller than that in the five-icon condition in a circular search array of a given diameter. This implies that icon search performance was significantly affected by the set size of a search array; thus, the larger the number of icons presented in a search array becomes, the more search time is needed to locate a target.

The findings of the present study showed that delay time affected search performance, indicating that a shorter delay time resulted in a shorter search time and higher accuracy. This result was similar to our expectation and consistent with the results from several previous studies.^{24,27} One possibility is that subjects needed to maintain the physical characteristics of the target icon in short-term memory and then accurately retrieve the target icon to search for it in a stimulus array. According to decay theory, an original piece of information gradually disappears unless something occurs to keep it intact. However, subjects could not rehearse the stimulus icons because they were placed in an iconic pattern during the delay time, and they were not semantic or phonetic stimuli. Subjects may have forgotten the physical characteristics of the target icon as time passed, leading to a decrease in subjects' search performance with an increase in the time delay. However, the present result was contrary to the results of Berry and Carpenter,³² who examined the effect of four different delay periods (15–60 min) on recall of the ROCF and found no differences between immediate and delayed recall trials. One possible reason for this discrepancy is the difference between studies in

icon complexity and the ROCF presented. Furthermore, the maximum delay time for our study was 5 sec, whereas the minimum delay time was 15 min in their study. A subject's recall performance might be significantly reduced regardless of delay if they were asked to maintain more complex figures during a longer period, and this might explain differences between our results and the findings of Berry and Carpenter.

No significant effect of visual field on icon search performance was found in the present study. This result was inconsistent with the findings of previous studies,^{15,36,37} which found right-visual-field superiorities for smaller displays and left-visual-field superiorities for larger displays across different processing tasks. One possible reason for the inconsistency lies in differences between the current study and previous studies in the task characteristics. For example, Pedersen and Polich¹⁵ assessed the hemispheric processing difference by presenting square matrices that varied in size and the number of filled-in cells and asked subjects to determine whether each stimulus matrix contained an odd or an even number of filled cells, a variation which did not change luminance levels. However, in the present study, subjects were asked to search for a target icon in a circular array presented on the right or left sides of the screen. Subjects needed to recognize, discriminate, and search for the icons and were then required to move the mouse and click on the target. However, in the former study, subjects only counted filled-in cells in the stimulus matrix. Hence, the task in the present study required more attentional resources and processing time than did that in the Pedersen and Polich study, resulting in a reduction of the effect of visual field on search performance in the current study.

Exposure time of a target icon did not significantly affect search performance. These results were inconsistent with those of previous studies,^{13,15,17,38} which found that the functions for recognition judgments exhibited different temporal dynamics over brief exposure durations (shorter than 100 msec) and that target icon memory (recognition) was possible at very brief exposures and was slightly influenced by increasing duration. Although this implies that the target icon had a longer exposure, participants could memorize the target icon, which allowed participants to search for a target in a search array, and search performance might be slightly influenced by increases in exposure duration. In the present study, however, exposure duration was at least 1 sec. Such an exposure duration may be enough for subjects to accurately memorize the target icons. Therefore, the search performance did not improve with increasing exposure duration, resulting in similar search performances for exposure durations of 1, 2, and 3 sec.

Our results have implications for the areas of human-machine-interface design, ranging from applications controlling industrial processes to mobile applications used by novices. For example, first, the findings can be applied to the design of effective Web sites, assuming that it is easy to

search for a target icon. Second, a suitable number and location of icons presented on a Web site can help observers select appropriate icons in a search array as quickly and accurately as possible. These results will make designers aware of the characteristics that are likely to be effective when they design a Web page or display. Unfortunately, subjects' field of attention did not affect search performance, a finding that was unexpected and contrary to previous studies. Therefore, further investigation of the effect of visual field on various stimuli and task characteristics is needed.

Acknowledgment

The authors thank Wan-Li Liao for her help in conducting parts of the experiments.

References

- W. Horton, *The Icon Book: Visual Symbols for Computer Systems and Documentation* (Wiley, New York, 1994).
- M. C. Boschman and J. A. J. Roufs, "Text quality metrics for visual display units: II. An experimental survey," *Displays* **18**, 45–64 (1997).
- R. Nasanen *et al.*, "Effects of stimulus contrast on performance and eye movements in visual search," *Vision Res.* **41**, 1817–1824 (2001).
- R. Nasanen *et al.*, "Display quality and the speed of visual letter search," *Displays* **22**, 107–113 (2001).
- S. McDougall *et al.*, "Searching for signs, symbols, and icons: Effects of time of day, visual complexity, and grouping," *J. Exptl. Psychol.: Appl.* **12**(2), 118–128 (2006).
- P. Verghese and S. P. McKee, "Visual search in clutter," *Vision Res.* **44**, 1217–1225 (2004).
- J. G. Seamon *et al.*, "Affective discrimination of stimuli that are not recognized: Effects of shadowing, masking, and cerebral laterality," *J. Exptl. Psychol.: Learning, Memory, and Cognition* **9**, 544–555 (1983).
- J. E. Jonsson and J. B. Hellige, "Lateralized effects of blurring: a test of the visual spatial frequency model of cerebral hemisphere asymmetry," *Neuropsychologia* **24**, 351–362 (1986).
- H. K. Pedersen and J. Polich, "Hemispheric differences for visual matrix processing: Stimulus size and spatial frequency effects," *Brain and Cognition* **47**, 525–538 (2001).
- A. A. Harrison, "Mere exposure," L. Berkowitz (ed.), *Advances in Experimental Social Psychology* (Academic Press, New York, 1997), pp. 39–83.
- P. N. Shapiro and S. Penrod, "Meta-analysis of facial identification studies," *Psychol. Bull.* **100**, 139–156 (1986).
- J. K. Reynolds and K. Pezdek, "Face recognition memory: The effects of exposure duration and encoding instruction," *Appl. Cognitive Psychol.* **6**, 279–292 (1992).
- C. Bundesen and L. Harms, "Single-letter recognition as a function of exposure duration," *Psychol. Res.* **62**, 275–279 (1999).
- J. E. Hoffman, "Interaction between global and local levels of a form," *J. Exptl. Psychol.: Human Perception and Performance* **6**, 222–234 (1980).
- L. Paquet and P. M. Merikle, "Global precedence: The effect of exposure duration," *Can. J. Psychol.* **38**(1), 45–53 (1984).
- D. Navon, "Forest before trees: The precedence of global features in visual perception," *Cognitive Psychol.* **9**, 353–383 (1977).
- J. G. Seamon *et al.*, "Critical importance of exposure duration for affective discrimination of stimuli that are not recognized," *J. Exptl. Psychol.: Learning, Memory, and Cognition* **10**(3), 465–469 (1984).
- B. N. S. Vlaskamp *et al.*, "Saccadic search performance: The effect of element spacing," *Exptl. Brain Res.* **167**(2), 246–259 (2005).
- T. Lindberg and R. Nasanen, "The effect of icon spacing and size on the speed of icon processing in the human visual system," *Displays* **24**, 111–120 (2003).
- D. G. Elmes *et al.*, "What is the role of affect in the spacing effect?" *Memory & Cognition* **11**, 144–151 (1983).
- N. Lavie *et al.*, "Load theory of selective attention and cognitive control," *J. Exptl. Psychol.: General* **133**(3), 339–354 (2004).
- G. D. Logan, "Attention in character-classification tasks: evidence for the automaticity of component stages," *J. Exptl. Psychol.: General* **107**, 32–63 (1978).
- A. V. Popple *et al.*, "Two sources of error in pop-out localization," *Vision Res.* **46**, 293–298 (2006).
- J. Podd, "The effects of memory load and delay on facial recognition," *Applied Cognitive Psychol.* **4**, 47–59 (1990).
- T. Hanari, "Information on location in visual persistence: Effects of temporal delay and duration of exposure in report of dot location," *Perceptual and Motor Skills* **80**, 643–650 (1995).
- B. A. Doshier, "Item interference and time delays in working memory: immediate serial recall," *Intl. J. Psychol.* **34**(5/6), 276–284 (1999).
- M. Singer and J. T. Wixted, "Effect of delay on recognition decisions: Evidence for a criterion shift," *Memory & Cognition* **34**(1), 125–137 (2006).
- P. Barkowitz and J. C. Brigham, "Recognition of faces: Own-race bias, incentive and time delay," *J. Appl. Soc. Psychol.* **12**, 255–268 (1982).
- K. R. Laughery, "Time delay and similarity effects in facial recognition," *J. Appl. Psychol.* **59**(4), 490–496 (1974).
- A. D. Yarmey, "Recognition memory for familiar public faces: effects of orientation and delay," *Psychonomic Sci.* **24**, 286–288 (1971).
- A. G. Goldstein and J. E. Chance, "Visual recognition memory for complex configurations," *Perception & Psychophysics* **9**, 237–241 (1970).
- D. T. R. and G. S. Carpenter, "Effects of four different delay periods on recall of the Rey–Osterrieth complex figure by older persons," *The Clinical Neuropsychologist* **6**(1), 80–84 (1992).
- O. H. MacLin *et al.*, "Race, arousal, attention, exposure, and delay," *Psychology, Public Policy, and Law* **7**(1), 134–152 (2001).
- A. M. Trisman and G. Gelade, "A feature-integration theory of attention," *Cognitive Psychol.* **12**, 97–136 (1980).
- K. Oberauer, "Selective attention to elements in working memory," *Exptl. Psychol.* **50**(4), 257–269 (2003).
- S. D. Christman *et al.*, "Hemispheric asymmetry in the processing of absolute versus relative spatial frequency," *Brain and Cognition* **16**, 62–73 (1991).
- J. Polich and C. D. Morgan, "Handedness and hemispheric differences for feature perturbations," *Brain and Cognition* **25**, 220–234 (1994).
- K. C. Huang, "Effects of computer icons and figure/background area ratios and color combinations on visual search performance on an LCD monitor," *Displays* **29**, 237–242 (2008).



Kuo-Chen Huang received his Doctoral degree in industrial management from National Taiwan University of Science and Technology in 2000. He has been a faculty member in the Department of Product Design at Ming Chuan University (Taiwan) since August 2004. His research activities are the area of ergonomic design and visual perception, focusing on legibility of Chinese text, visual performance, and human–computer interaction.



Wen-Te Chang received his Doctoral degree from the Design Graduate School of National Taiwan University of Science and Technology in 2009. He has been a faculty member in the Department of Product Design at Ming Chuan University (Taiwan) since 1998. His research activities are in the area of Wayfinding, human–computer interaction design, and design research.



Wan-Li Wei is currently an Assistant Professor in the Department of Product Design at Ming Chuan University. He received his Ph.D. in 2006 for research on multiple-criteria decision analysis from the Graduate School of Design at National Taiwan University of Science and Technology (NTUST). His research specialties are new product development, interface design, visual communication design, and design methodology.